

Sprint 2 — Budget One Pager

CS 5388 / CS 4390 Software Project Management

Team: 1 Project: Damn Parking Date: October 5, 2026

Proposed build funding: \$35,469, plus \$6,000 per operating year. This is an approval request, not approved funding or actual spending. New allocations, fees, risk probabilities and reserves below are illustrative assumptions.

1 Cost assumptions and accountable roles

Cost-driving assumption	Why it affects cost	Owner
Six-story slice; larger launch deferred	Limits features, locations and rollout	Project manager
Existing cameras and occupancy feed	Avoids installation; access fees remain possible	Sponsor / technical lead
4 people × 20 productive hours/week	Sets capacity and coordination effort	Project manager
\$500 build services; \$500/month operations	Data, maps, hosting and support costs	Technical lead

2 Effort and staffing

Function Points: 340–640 hours; Story Points: 360–680. PERT: 538 hours = 4.138 FTE-months at 130 productive hours/month. Four people at 20 hours/week give 6.73 weeks. Loaded labor: \$60/productive hour = \$7,800/FTE-month.

Role	M1 hours	M2 hours	Total hours	FTE-months
Developer	200	123	323	2.485
Quality assurance	50	40	90	0.692
DevOps / data	40	20	60	0.462
Project manager	30	35	65	0.500
Total	320	218	538	4.138

Role allocation reconciles the team worksheet to 538 hours: 33 previously unassigned hours go to development. FTE-months = hours ÷ 130 (rounded independently). Month 1 uses four weeks; Month 2 uses 2.725. Coordination and integration are included, not added again.

3 Build cost and annual operations

Cost line	Calculation / basis	Amount
Loaded labor	538 hours × \$60	\$32,280
Build services / tools	2 months × \$250 allowance	\$500
Base build estimate	Labor + non-labor	\$32,780
Contingency reserve	Named risks below; PM controlled	\$1,000
Proposed cost baseline	Base + contingency	\$33,780
Management reserve	5% of baseline; sponsor controlled	\$1,689
Total build request	Baseline + management reserve	\$35,469
Annual operating cost	Hosting, data, maps, support: \$500 × 12	\$6,000

4 Risk reserves and spending plan

Named fee risks: data access $40\% \times \$1,500 = \600 ; map upgrade $30\% \times \$1,000 = \300 ; test service $20\% \times \$500 = \100 . EMV totals \$1,000; development rework is already in the effort range. The assumed 5% management reserve covers unforeseen costs outside the baseline.

Base spend: M1 \$19,450; M2 \$13,330; total \$32,780. Reserves are held until needed. Labor share: 98.5%; method midpoints differ 12.7%; no sponsor cap is recorded. First-year funding: \$41,469 including reserves and operations. Earlier ROI uses \$38,280 and excludes the new build allowance and reserves.